DBMS BASIC

1). What is Database?

> Database is a collection of related data and data is a collection of facts and figures that can be processed to produce information.

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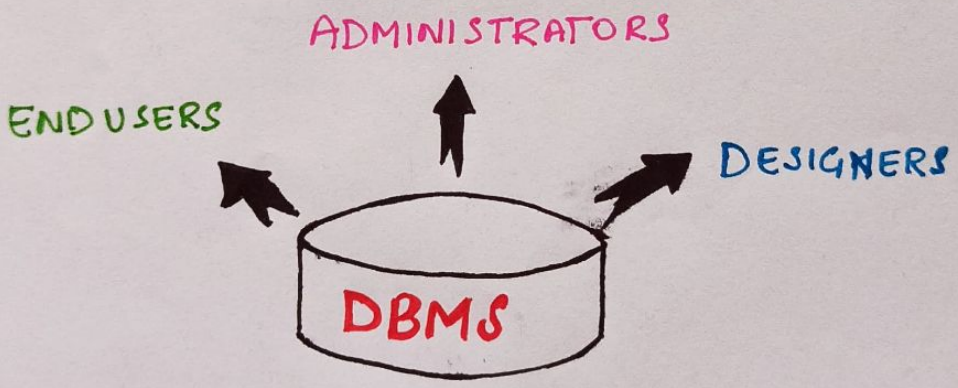
2). What is DBMS?

> A DataBase Management system (DBMS) stores the data in a such a way that it becomes easier to retrive, manipulate and produce information.

3). The user of a DBMS can be broadly categorized as :

>



- END USERS :-

  END users are those who actually reap the benefits of having a DBMS.

- Administration :-

  Administer maintain the DBMS and are responsible for administrating the database.

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- **Designers :-**

Designers are the group of people who actually work on the designing part of the database.

4. DBMS Architecture

> The design of a DBMS depends on its architecture. It can be centralized or decentralized or hierarchical.

The architecture of a DBMS can be seen as a either single tier or multitier.

In 1 - tier Architecture, the DBMS is the only entity where the user directly sits on the DBMS and uses it. Any changes done here will directly be done on the DBMS itself.

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If the Architecture of DBMS is 2 - tier, then it must have an application through which the DBMS can be accessed. programmers use 2-tier architecture where they access the DBMS by means of an application.

A 3 - tier Architecture seprates its tiers from each other based on the complexity of the users and how they use the data present in the database. It is most widely used architecture to design to a DBMS.

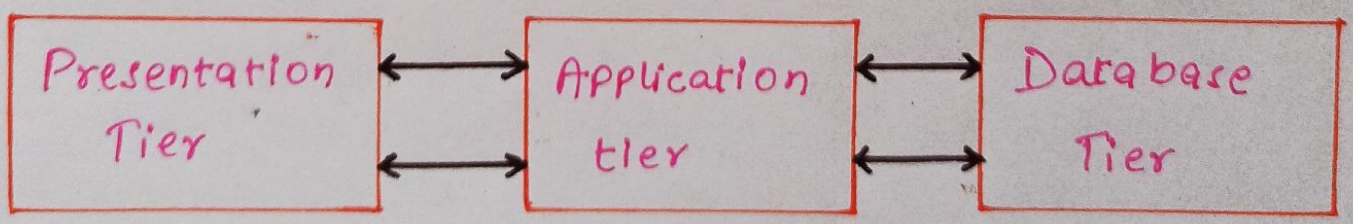

Fig 1.2 (Architecture of DBMS) 3 tier.

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1. Database (Data) Tier :- At this tier, the database resides along with its query processing languages. We also have the relations that define the data and their constrains at this level.

2. Application (Middle) Tier :- At this tier, reside the application server and programs that access the database for a user, this application tier presents an abstracted view of the database. End users are unware of any existance of the database beyond the application.

3. User (presentation) Tier :- END users operate on this tier and they know nothing about any existance of the database beyond this layer.

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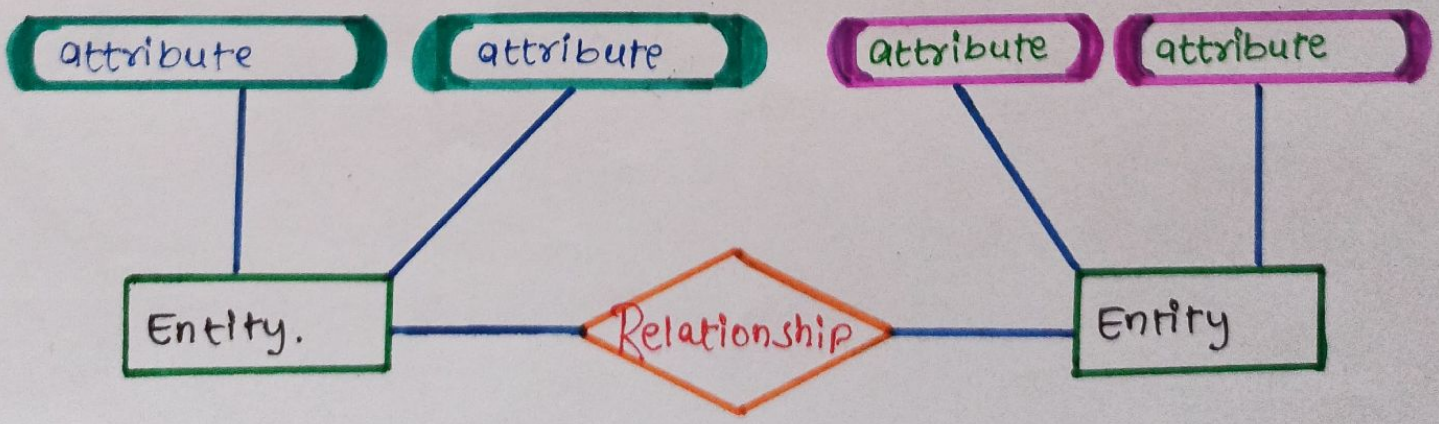
5). What is ER Model?

> Entity - Relationship (ER) model is based on the notion of real world entities and relationships among them. ER model is best used for the conceptual design of a database.

ER model is based on:

- Entities and their attributes.
- Relationships among entities.



- Entity :- An entity of an ER-model is a real-world entity having properties called attributes.
  Every attribute is defined by the set of values called domain.

Example :

database - student (school database) where
Entity - student
Attribute - name, age, class, etc.

Relationship :-

The logical association among entities is called Relationship. Relationships are mapped with entities in various ways.

Mapping cordinaties :-

- One to one
- One to many
- many to one
- many to many

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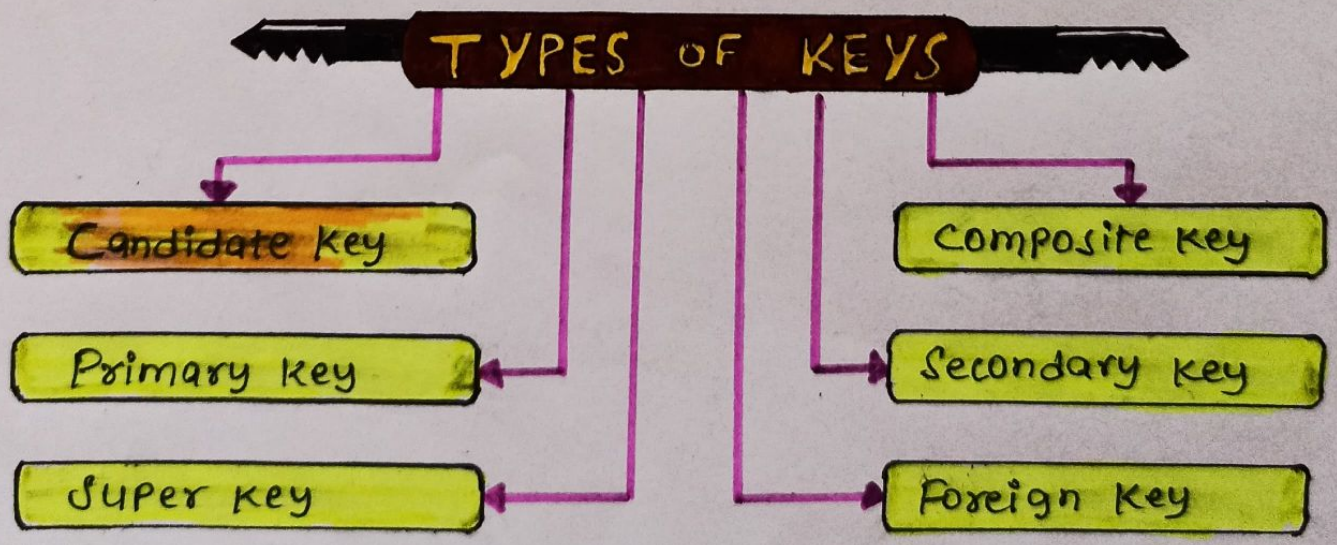
6). Types of Keys

There are various types of keys in a relationship which are:-



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- **Candidate key :-** The candidate key in a table are defined as the set of keys that is minimal and can uniquely identify any data row in the table.

  Example :- Studid, Roll NO, and email are candidate keys.

- **Primary key :-** The primary key is selected from one of the candidate keys and becomes the identifying key of a table. It can uniquely identify any data row of the table.

  Example :- STUD_NO, as well as STUD_PHONE, are candidate keys for relation STUDENT but STUD_NO can be chosen as the primary (only one out of many candidate keys).

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- **Super key :-** Super key is the superset of primary key. The super key contains a set of attributes, including the primary key, which can uniquely identify any data row in the table.

  Example :- STUD_NO, (STUD_NO, STUD_NAME), etc.

- **Composite key :-** If any single attribute of a table is not capable of being the key i.e it cannot identify a row uniquely, then we combine two or more attributes to form a key. This is known as composite key.

  Example :- FULLNAME + DOB can be combined together to access the details of a student.

- **Secondary key :-** Only one of the candidate key is selected as the primary key. The rest of them are known as secondary key.

  Example :- STUD_NO, STUD_PHONE.

- Foreign key :- A foreign key is an attribute value in a table that acts as the primary key in another table. Hence the foreign key is useful in linking together two tables. Data should be entered in the foreign key column with great care, as wrong entered data can invalidate the relationship between the two tables.

Example :- STUD_NO in STUDENT_COURSE is a foreign key to STUD_NO in student relation.

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7 Operators :

① . Select :- The SELECT statement in SQL is used is to retrieve or fatch data from a database. The data return is stored in the table called as result table. It is also called result-set.

Syntax :-

SELECT column 1, Column 2, from table-name
{ column 1, column 2 : names of field of table .
table_name : from where we want to fetch }.

② From :- From clause can be used to specify a sub-query expression in SQL. The relation produced by the sub-query is then used as a new relation on which outer query is applied.

Syntax :-

SELECT Column1, column 2 from
(SELECT column x , as C1, column -y FROM table
WHERE PREDICATE_X)
WHERE PREDICATE 3

③. **WHERE**:- WHERE keyword is used for fetching filtered data in a result set.

- It is used to fetch data according to a particular criteria.
- WHERE keyword can also be used to filter data by matching patterns.

Syntax:-
```
SELECT column 1, column 2 FROM
table_name WHERE column_name
operator value;
```

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④. EXISTS:- The EXISTS condition in SQL is used to check whether the result of a related nested query is empty or not.

Syntax:-
```
SELECT column_name (s)
FROM table_name
WHERE EXISTS
  (SELECT column_name (s))
    FROM table_name
    WHERE condition);
```

⑤ **GROUP BY**:- The GROUP BY statement in SQL is used to arrange identical data into groups with the help of some functions.

Syntax:-
```
SELECT column 1, function_name (column 2).
FROM table_name
WHERE condition
GROUP BY column 1, column 2.
ORDER BY column 1, column 2;
```

(6). **ORDER BY** :- The ORDER BY keyword is used to sort the result-set in ascending or descending order. The ORDER BY keyword sorts the records in ascending order by defaut. To sort the record in descending order, use the DESE keyword.

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Syntax :-

```
SELECT column1, column2, ...
FROM table_name
ORDER BY column1, column2, ...
ASC|DESC;
```